

Meetings 3–4 — Multivariable Integrals

Mathematics for Chemists | 2×2.5 hours

Meeting 3

Part 1 — Change of Coordinates (~45 min)

1.1 Coordinate Systems as Smooth Maps

A coordinate system is simply a smooth map $\Phi: U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^n$ that assigns to each set of coordinates $(u_1, \dots, u_n)$ a point $(x_1, \dots, x_n)$ in space. For the map to be useful, it must be **invertible**: given a point in space, we should be able to recover its coordinates uniquely.

The fundamental question of this part is: *when is Φ locally invertible?* The answer requires computing the **Jacobian matrix** $D\Phi$ — the matrix of all partial derivatives of Φ — and checking whether its determinant is nonzero. We will see why momentarily.

1.2 Polar Coordinates

The polar coordinate map $\Phi: (0, \infty) \times [0, 2\pi) \rightarrow \mathbb{R}^2 \setminus \{0\}$ is:

$$x = r \cos \theta, \quad y = r \sin \theta.$$

Its Jacobian matrix is the matrix of partial derivatives of (x, y) with respect to (r, θ) :

$$D\Phi = \begin{pmatrix} \partial x / \partial r & \partial x / \partial \theta \\ \partial y / \partial r & \partial y / \partial \theta \end{pmatrix} = \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix}.$$

Its determinant (the **Jacobian determinant**) is:

$$J = \det D\Phi = r \cos^2 \theta + r \sin^2 \theta = r.$$

1.3 When Is the Map 1-to-1? The Role of the Jacobian

Recall from the discussion of line integrals: a smooth map $\Phi: U \rightarrow \mathbb{R}^n$ is **locally invertible** near a point $p \in U$ if and only if $\det D\Phi(p) \neq 0$. This is the **Inverse Function Theorem** — a consequence of the Implicit Function Theorem we use throughout the course.

Inverse Function Theorem

Let $\Phi: U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^n$ be continuously differentiable. If $\det D\Phi(p) \neq 0$ at some point $p \in U$, then Φ is a smooth invertible map in some neighbourhood of p , and

$$D(\Phi^{-1})(\Phi(p)) = (D\Phi(p))^{-1}.$$

Polar coordinates. $J = r$, so the map is locally invertible everywhere except at $r = 0$. At the origin, polar coordinates *degenerate*: the angle θ is undefined, and every circle of radius $r \rightarrow 0$ collapses to a single point. This is not a pathology of the function being described — it is a failure of the coordinate system itself.

Draw on board: the polar grid collapsing to a point at $r = 0$. Contrast with the Cartesian grid, which is uniform everywhere.

When does the map fail to be 1-to-1 globally? Even if $J \neq 0$ everywhere, a map may fail to be globally 1-to-1. Polar coordinates: the ray $\theta = 0$ and the ray $\theta = 2\pi$ are the same ray, so points with $\theta = 0$ and $\theta = 2\pi$ map to the same (x, y) . This is why the domain is restricted to $\theta \in [0, 2\pi)$. The Jacobian detects *local* invertibility only; global invertibility requires additional analysis of the boundary of the domain.

Summary of criteria.

Condition	Meaning
$\det D\Phi(p) \neq 0$	Φ is locally invertible near p
$\det D\Phi(p) = 0$	Φ is not locally invertible; coordinate singularity
Φ injective on U and $\det D\Phi \neq 0$ on U	Φ is globally 1-to-1 on U

1.4 Cylindrical Coordinates

The cylindrical coordinate map $\Phi: (0, \infty) \times [0, 2\pi) \times \mathbb{R} \rightarrow \mathbb{R}^3 \setminus \{z\text{-axis}\}$:

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z.$$

Jacobian matrix:

$$D\Phi = \begin{pmatrix} \cos \theta & -r \sin \theta & 0 \\ \sin \theta & r \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad J = r.$$

The map degenerates along the entire z -axis ($r = 0$). Natural for: linear molecules, cylindrically symmetric charge distributions, NMR problems.

1.5 Spherical Coordinates

The spherical coordinate map $\Phi: (0, \infty) \times (0, \pi) \times [0, 2\pi) \rightarrow \mathbb{R}^3 \setminus \{\text{poles and } z\text{-axis}\}$:

$$x = r \sin \theta \cos \varphi, \quad y = r \sin \theta \sin \varphi, \quad z = r \cos \theta.$$

Here $r > 0$, $\theta \in (0, \pi)$ (polar angle from z -axis), $\varphi \in [0, 2\pi)$ (azimuthal angle).

Jacobian matrix:

$$D\Phi = \begin{pmatrix} \sin \theta \cos \varphi & r \cos \theta \cos \varphi & -r \sin \theta \sin \varphi \\ \sin \theta \sin \varphi & r \cos \theta \sin \varphi & r \sin \theta \cos \varphi \\ \cos \theta & -r \sin \theta & 0 \end{pmatrix}.$$

Expanding (see Exercise 2):

$$J = \det D\Phi = r^2 \sin \theta.$$

The map degenerates at $r = 0$ (origin) and at $\theta = 0, \pi$ (the poles, where φ is undefined). Natural for: atoms, molecules with spherical symmetry, all of quantum chemistry.

Convention warning: physicists and mathematicians often swap θ and φ . The convention above (θ polar, φ azimuthal) is standard in quantum chemistry.

System	Coordinates	Transformation	Jacobian	Degeneracy
Polar	(r, θ)	$x = r \cos \theta, y = r \sin \theta$	r	$r = 0$
Cylindrical	(r, θ, z)	$x = r \cos \theta, y = r \sin \theta, z = z$	r	$r = 0$
Spherical	(r, θ, φ)	$x = r \sin \theta \cos \varphi, \dots$	$r^2 \sin \theta$	$r = 0; \theta = 0, \pi$

1.6 Exercises

Exercise 1. Describe the following sets in Cartesian coordinates.

- (a) Polar: $\{(r, \theta) : r \leq 1, 0 \leq \theta \leq \pi/2\}$
- (b) Cylindrical: $\{(r, \theta, z) : r = 2, 0 \leq z \leq 3\}$
- (c) Spherical: $\{(r, \theta, \varphi) : r = R, \theta \in (0, \pi), \varphi \in [0, 2\pi)\}$

Answers: (a) quarter disk $x^2 + y^2 \leq 1, x, y \geq 0$; (b) the cylinder $x^2 + y^2 = 4, 0 \leq z \leq 3$; (c) the sphere of radius R (excluding poles).

Exercise 2. Verify that the Jacobian of the spherical coordinate map equals $r^2 \sin \theta$ by expanding the 3×3 determinant.

Hint: expand along the third column.

Exercise 3. The map $\Phi(u, v) = (u^2 - v^2, 2uv)$ sends $\mathbb{R}^2$ to $\mathbb{R}^2$.

- (a) Compute $D\Phi$ and $J = \det D\Phi$.
- (b) At which points is Φ not locally invertible?
- (c) Show that $\Phi(u, v) = \Phi(-u, -v)$ for all (u, v) : the map is never globally 1-to-1 on all of $\mathbb{R}^2$.

(Note: this is the squaring map $z \mapsto z^2$ in disguise, with $z = u + iv$.)

Answer: $J = 4(u^2 + v^2)$; fails at the origin only.

Exercise 4. Express the surface $z = \sqrt{x^2 + y^2}$ (a cone) in cylindrical and spherical coordinates. At which points does each coordinate system fail to be 1-to-1?

Answer: cylindrical: $z = r$ (the entire surface is covered, no issue except $r = 0$); spherical: $\theta = \pi/4$ (cone at 45°), $r > 0$.

Part 2 — Double Integrals (~45 min)

2.1 From 1D to 2D: Length, Area, Volume

Recall the 1D integral as a limit of Riemann sums. The most basic fact: if $f = 1$,

$$\int_a^b 1 \, dx = b - a = \text{length of } [a, b].$$

Moving to 2D: partition a rectangle $[a, b] \times [c, d]$ into small rectangles of area $\Delta x \Delta y$. The sum of all these areas is the area of the rectangle:

$$\iint_{[a,b] \times [c,d]} 1 \, dx \, dy = (b - a)(d - c) = \text{area}.$$

In 3D: partition a box into small boxes of volume $\Delta x \Delta y \Delta z$:

$$\iiint_{\Omega} 1 \, dx \, dy \, dz = \text{volume of } \Omega.$$

For a general region $D \subseteq \mathbb{R}^2$:

$$\iint_D dx \, dy = \text{area of } D, \quad \iiint_{\Omega} dx \, dy \, dz = \text{volume of } \Omega.$$

2.2 Computing Areas and Volumes in Cartesian Coordinates

For a region $D = \{(x, y) : a \leq x \leq b, g(x) \leq y \leq h(x)\}$:

$$\iint_D dx \, dy = \int_a^b \int_{g(x)}^{h(x)} dy \, dx = \int_a^b (h(x) - g(x)) \, dx.$$

This is the formula for the area between two curves — the same formula from first-year calculus, now interpreted as a double integral.

Example 1: area of a triangle. $D = \{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq x\}$:

$$\iint_D dx \, dy = \int_0^1 \int_0^x dy \, dx = \int_0^1 x \, dx = \frac{1}{2}. \quad \checkmark$$

Example 2: volume of a pyramid. $\Omega = \{(x, y, z) : x, y, z \geq 0, x + y + z \leq 1\}$:

$$\iiint_{\Omega} dx \, dy \, dz = \int_0^1 \int_0^{1-x} \int_0^{1-x-y} dz \, dy \, dx = \int_0^1 \int_0^{1-x} (1-x-y) \, dy \, dx = \int_0^1 \frac{(1-x)^2}{2} \, dx = \frac{1}{6}.$$

2.3 The Integral of $f(x, y)$ as Volume Under a Surface

If $f(x, y) \geq 0$, then $\iint_D f(x, y) \, dx \, dy$ is the **volume of the solid** lying between the region D in the xy -plane and the surface $z = f(x, y)$ above it.

This follows from the Riemann sum interpretation: each term $f(x_i, y_j) \Delta x \Delta y$ is the volume of a thin rectangular column of height $f(x_i, y_j)$ and base area $\Delta x \Delta y$.

Example 3: volume under a paraboloid. $f(x, y) = 1 - x^2 - y^2$, $D = \{(x, y) : x^2 + y^2 \leq 1\}$:

In Cartesian coordinates this is hard to set up. We will return to it using polar coordinates in Part 3. For now, note that $\iint_D (1 - x^2 - y^2) \, dx \, dy$ should give $\pi/2$ (by symmetry: the paraboloid reaches height 1 at the origin and 0 on the unit circle).

Example 4: probability (chemistry motivation). The Maxwell–Boltzmann speed distribution in 2D is $f(v_x, v_y) \propto e^{-(v_x^2 + v_y^2)/2k_B T/m}$. The probability of finding the speed in a disk of radius v_0 is:

$$P(|\mathbf{v}| \leq v_0) = C \iint_{v_x^2 + v_y^2 \leq v_0^2} e^{-(v_x^2 + v_y^2)/2k_B T/m} \, dv_x \, dv_y.$$

This integral is essentially the Gaussian integral in 2D, which we will evaluate exactly using polar coordinates and the famous trick of Meeting 4.

2.4 The Area of a Disk: Why We Need New Coordinates

Let $D = \{(x, y) : x^2 + y^2 \leq R^2\}$ be the disk of radius R . In Cartesian coordinates:

$$\iint_D dx dy = \int_{-R}^R \int_{-\sqrt{R^2-x^2}}^{\sqrt{R^2-x^2}} dy dx = \int_{-R}^R 2\sqrt{R^2-x^2} dx.$$

This requires a trigonometric substitution and is laborious. The answer is, of course, πR^2 , but the Cartesian route gives no insight into *why*.

In polar coordinates, the disk is simply $\{0 \leq r \leq R, 0 \leq \theta \leq 2\pi\}$ — a rectangle in (r, θ) -space. To use this, we need to know how areas transform under the coordinate change. The answer involves the Jacobian determinant, and we will derive it now.

Meeting 4

Part 3 — Change of Variables and Fubini's Theorem (~50 min)

3.1 The Change of Variables Formula

We hinted in Part 1 that the Jacobian determinant is the key to coordinate changes. Here is the precise statement.

Theorem (Change of Variables)

Let $\Phi: U \rightarrow V$ be a smooth bijection between open sets in $\mathbb{R}^n$ with $\det D\Phi \neq 0$ on U . Then for any integrable function $f: V \rightarrow \mathbb{R}$:

$$\int_V f(x) dx = \int_U f(\Phi(u)) |\det D\Phi(u)| du.$$

Why $|\det D\Phi|$? A small box $[u_0, u_0 + du_1] \times \cdots$ in u -space is sent by Φ to an approximate parallelepiped in x -space spanned by the columns $(\partial\Phi/\partial u_i) du_i$. The volume of this parallelepiped is $|\det D\Phi| du_1 \cdots du_n$. This is the geometric meaning of the Jacobian determinant: it measures how much Φ stretches or compresses volumes. At points where $|\det D\Phi| = 0$, the map collapses volume — which is exactly why coordinates degenerate there.

Area of a disk. With $\Phi(r, \theta) = (r \cos \theta, r \sin \theta)$, $|\det D\Phi| = r$:

$$\iint_D dx dy = \int_0^{2\pi} \int_0^R r dr d\theta = 2\pi \cdot \frac{R^2}{2} = \pi R^2. \quad \checkmark$$

The factor r converts the polar “rectangle” $dr d\theta$ into the true area element in the plane.

Volume of a ball. With spherical coordinates, $|\det D\Phi| = r^2 \sin \theta$:

$$\iiint_{|\mathbf{x}| \leq R} dx dy dz = \int_0^{2\pi} \int_0^\pi \int_0^R r^2 \sin \theta dr d\theta d\varphi = 2\pi \cdot 2 \cdot \frac{R^3}{3} = \frac{4\pi R^3}{3}. \quad \checkmark$$

3.2 Fubini's Theorem

Theorem (Fubini)

If $f: [a, b] \times [c, d] \rightarrow \mathbb{R}$ is continuous, then:

$$\iint_{[a,b] \times [c,d]} f(x, y) \, dx \, dy = \int_a^b \int_c^d f(x, y) \, dy \, dx = \int_c^d \int_a^b f(x, y) \, dx \, dy.$$

The order of integration can be freely exchanged.

Fubini's theorem extends to general regions (with appropriate hypotheses on f). Changing the order of integration can dramatically simplify a computation.

Example. $\int_0^1 \int_x^1 e^{y^2} \, dy \, dx$.

The inner integral $\int_x^1 e^{y^2} \, dy$ cannot be computed in closed form. Switching order: the region is $\{0 \leq x \leq y, 0 \leq y \leq 1\}$, so:

$$\int_0^1 \int_0^y e^{y^2} \, dx \, dy = \int_0^1 y e^{y^2} \, dy = \left[\frac{1}{2} e^{y^2} \right]_0^1 = \frac{1}{2}(e - 1).$$

3.3 Exercises

Exercise 5. Compute the following integrals using polar coordinates.

- (a) $\iint_D e^{-(x^2+y^2)} \, dx \, dy$ where $D = \{x^2 + y^2 \leq R^2\}$.
- (b) $\iint_D \sqrt{x^2 + y^2} \, dx \, dy$ where D is the unit disk.
- (c) $\iint_D (x^2 + y^2) \, dx \, dy$ where $D = \{1 \leq x^2 + y^2 \leq 4\}$ (annulus).

Answers: (a) $\pi(1 - e^{-R^2})$; (b) $2\pi/3$; (c) $\pi \int_1^2 r^2 \cdot r \, dr \cdot 2\pi \rightarrow \text{wait, } \int_0^{2\pi} d\theta \int_1^2 r^2 \cdot r \, dr = 2\pi \cdot [r^4/4]_1^2 = 2\pi \cdot 15/4 = 15\pi/2$.

Exercise 6. Compute the volume under the paraboloid $z = 1 - x^2 - y^2$ above the unit disk $D = \{x^2 + y^2 \leq 1\}$.

Use polar coordinates: $\int_0^{2\pi} \int_0^1 (1 - r^2)r \, dr \, d\theta = 2\pi \int_0^1 (r - r^3) \, dr = 2\pi \cdot [r^2/2 - r^4/4]_0^1 = 2\pi \cdot 1/4 = \pi/2$.

Exercise 7. Compute $\iint_D xy \, dx \, dy$ where $D = \{(x, y) : x^2 + y^2 \leq 1, x \geq 0, y \geq 0\}$ (first quadrant of unit disk).

In polar: $\int_0^{\pi/2} \int_0^1 r \cos \theta \cdot r \sin \theta \cdot r \, dr \, d\theta = \int_0^{\pi/2} \sin \theta \cos \theta \, d\theta \cdot \int_0^1 r^3 \, dr = \frac{1}{2} \cdot \frac{1}{4} = \frac{1}{8}$.

Exercise 8. Switch the order of integration and evaluate:

$$\int_0^1 \int_y^1 \sin(\pi x^2) \, dx \, dy.$$

Region: $0 \leq y \leq x \leq 1$. *Switching:* $\int_0^1 \int_0^x \sin(\pi x^2) \, dy \, dx = \int_0^1 x \sin(\pi x^2) \, dx = [-\cos(\pi x^2)/(2\pi)]_0^1 = 1/(\pi)$.

Exercise 9 (chemistry). The electron density of the hydrogen 1s orbital is $\rho(r) = \frac{1}{\pi a_0^3} e^{-2r/a_0}$. Verify the normalisation $\int_0^\infty \int_0^\pi \int_0^{2\pi} \rho(r) r^2 \sin \theta d\varphi d\theta dr = 1$.

Angular integral: 4π . *Radial:* $\frac{4}{a_0^3} \int_0^\infty r^2 e^{-2r/a_0} dr = \frac{4}{a_0^3} \cdot 2! \cdot (a_0/2)^3 = \frac{4}{a_0^3} \cdot \frac{a_0^3}{4} = 1$.

Part 4 — Classic Results (~40 min)

4.1 The Gaussian Integral

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}.$$

Proof via double integral and polar coordinates. Let $I = \int_{-\infty}^{\infty} e^{-x^2} dx$. Consider I^2 :

$$I^2 = \left(\int_{-\infty}^{\infty} e^{-x^2} dx \right) \left(\int_{-\infty}^{\infty} e^{-y^2} dy \right) = \iint_{\mathbb{R}^2} e^{-(x^2+y^2)} dx dy.$$

Converting to polar coordinates ($r \in [0, \infty)$, $\theta \in [0, 2\pi)$):

$$I^2 = \int_0^{2\pi} \int_0^\infty e^{-r^2} r dr d\theta = 2\pi \int_0^\infty r e^{-r^2} dr = 2\pi \cdot \left[-\frac{1}{2} e^{-r^2} \right]_0^\infty = 2\pi \cdot \frac{1}{2} = \pi.$$

Therefore $I = \sqrt{\pi}$. $\square$

Why this matters in chemistry. The normalised Gaussian $e^{-x^2/2}/\sqrt{2\pi}$ is the probability density of the normal distribution. Every measurement uncertainty is modelled by a Gaussian; every spectral line has a Gaussian profile. The factor $\sqrt{\pi}$ that appears throughout quantum chemistry — in the normalisation of harmonic oscillator eigenfunctions, in the error function, in the Debye–Waller factor — originates here.

4.2 The Basel Problem: $\zeta(2) = \pi^2/6$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = 1 + \frac{1}{4} + \frac{1}{9} + \cdots = \frac{\pi^2}{6}.$$

This was proved by Euler in 1734 and shocked the mathematical community. Here is a proof using double integrals.

Step 1. Use the geometric series: for $0 \leq xy < 1$, $\frac{1}{1-xy} = \sum_{n=0}^{\infty} (xy)^n$. Integrating over the unit square $(0, 1)^2$:

$$\iint_{(0,1)^2} \frac{dx dy}{1-xy} = \sum_{n=0}^{\infty} \iint_{(0,1)^2} (xy)^n dx dy = \sum_{n=0}^{\infty} \frac{1}{(n+1)^2} = \sum_{n=1}^{\infty} \frac{1}{n^2}.$$

Step 2: change of variables. Apply the 45° rotation $u = \frac{x+y}{\sqrt{2}}$, $v = \frac{x-y}{\sqrt{2}}$, with inverse $x = \frac{u+v}{\sqrt{2}}$, $y = \frac{u-v}{\sqrt{2}}$. The Jacobian of a rotation is 1, so $dx dy = du dv$. Also:

$$xy = \frac{(u+v)(u-v)}{2} = \frac{u^2 - v^2}{2}, \quad 1 - xy = 1 - \frac{u^2 - v^2}{2} = \frac{2 - u^2 + v^2}{2}.$$

The unit square $(0,1)^2$ in (x,y) maps to a square rotated 45° in (u,v) , with vertices at $(0,0) \rightarrow (0,0)$, $(1,0) \rightarrow (1/\sqrt{2}, -1/\sqrt{2})$, $(0,1) \rightarrow (1/\sqrt{2}, 1/\sqrt{2})$, $(1,1) \rightarrow (\sqrt{2}, 0)$. By the symmetry $v \mapsto -v$ (which leaves the integrand unchanged), the integral over the full rotated square equals twice the integral over the upper half, which splits into two triangles:

$$T_1 = \left\{ 0 \leq v \leq u \leq \frac{1}{\sqrt{2}} \right\}, \quad T_2 = \left\{ 0 \leq v \leq \frac{1}{\sqrt{2}} - u, \frac{1}{\sqrt{2}} \leq u \leq \sqrt{2} \right\}.$$

By the further symmetry $u \mapsto \sqrt{2} - u$ of T_2 relative to T_1 , both triangles give the same integral, so:

$$\iint_{(0,1)^2} \frac{dx dy}{1 - xy} = 4 \iint_{T_1} \frac{du dv}{1 - \frac{u^2 - v^2}{2}} = 4 \int_0^{1/\sqrt{2}} \int_0^u \frac{2 dv du}{2 - u^2 + v^2}.$$

Step 3: the inner integral. For fixed u , integrate over $v \in [0, u]$:

$$\int_0^u \frac{2 dv}{2 - u^2 + v^2}.$$

Write $a^2 = 2 - u^2$ (valid for $u < \sqrt{2}$, which holds on T_1 since $u \leq 1/\sqrt{2}$). Using $\int dv/(a^2 + v^2) = \frac{1}{a} \arctan(v/a) + C$:

$$\int_0^u \frac{2 dv}{a^2 + v^2} = \frac{2}{a} \arctan\left(\frac{v}{a}\right) \Big|_0^u = \frac{2}{\sqrt{2 - u^2}} \arctan\left(\frac{u}{\sqrt{2 - u^2}}\right).$$

Step 4: recognise the arctan. Substitute $u = \sqrt{2} \sin \phi$, so $\sqrt{2 - u^2} = \sqrt{2} \cos \phi$ and $u/\sqrt{2 - u^2} = \tan \phi$. Thus $\arctan(u/\sqrt{2 - u^2}) = \phi = \arcsin(u/\sqrt{2})$. At $u = 0$: $\phi = 0$. At $u = 1/\sqrt{2}$: $\phi = \pi/4$.

The inner integral becomes $\frac{2}{\sqrt{2 - u^2}} \arcsin\left(\frac{u}{\sqrt{2}}\right)$.

Step 5: the outer integral.

$$\frac{\pi^2}{6} \stackrel{?}{=} 4 \int_0^{1/\sqrt{2}} \frac{2}{\sqrt{2 - u^2}} \arcsin\left(\frac{u}{\sqrt{2}}\right) du.$$

Substitute $u = \sqrt{2} \sin \phi$, $du = \sqrt{2} \cos \phi d\phi$, $\sqrt{2 - u^2} = \sqrt{2} \cos \phi$; limits $0 \rightarrow 0$, $1/\sqrt{2} \rightarrow \pi/4$:

$$4 \int_0^{\pi/4} \frac{2}{\sqrt{2} \cos \phi} \cdot \phi \cdot \sqrt{2} \cos \phi d\phi = 4 \int_0^{\pi/4} 2\phi d\phi = 8 \left[\frac{\phi^2}{2} \right]_0^{\pi/4} = 4 \cdot \frac{\pi^2}{16} = \frac{\pi^2}{4}.$$

Wait — this gives $\pi^2/4$, not $\pi^2/6$. Recount: the symmetry argument above contributes a factor of 4 (factor 2 for $v \mapsto -v$, factor 2 for the two triangles T_1 and T_2), but we also have the factor of 2 from $\frac{2 dv}{2 - u^2 + v^2}$, giving overall factor 8 on T_1 . Let us redo the splitting carefully.

The rotated square has vertices $(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$, $(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}})$, $(\sqrt{2}, 0)$, $(0, 0)$. It consists of:

$$S^+ = \{0 \leq v \leq u, u + v \leq \sqrt{2}\}, \quad S^- = \{-u \leq v \leq 0, u - v \leq \sqrt{2}\}.$$

By $v \mapsto -v$ symmetry, $\iint_{S^+} = \iint_{S^-}$, so:

$$\iint_{(0,1)^2} \frac{dx dy}{1 - xy} = 2 \iint_{S^+} \frac{2 du dv}{2 - u^2 + v^2}.$$

The region S^+ further splits at $u = 1/\sqrt{2}$:

$$T_1 = \{0 \leq v \leq u \leq \frac{1}{\sqrt{2}}\}, \quad T_2 = \{0 \leq v \leq \sqrt{2} - u, \frac{1}{\sqrt{2}} \leq u \leq \sqrt{2}\}.$$

By the substitution $u \mapsto \sqrt{2} - u$, $v \mapsto v$ (which maps T_2 to T_1 and preserves $2 - u^2 + v^2 \mapsto 2 - (\sqrt{2} - u)^2 + v^2 = u^2 + 2\sqrt{2}u - v^2$) — actually T_1 and T_2 are *not* related by this simple symmetry. We integrate them separately.

On T_1 : (as computed above) $2 \int_0^{1/\sqrt{2}} \frac{2}{\sqrt{2-u^2}} \arcsin\left(\frac{u}{\sqrt{2}}\right) du = 2 \int_0^{\pi/4} 2\phi d\phi = 4 \cdot \frac{\pi^2}{32} = \frac{\pi^2}{8}.$

On T_2 : By the substitution $u = \sqrt{2} - s$, $v = v$ (so s runs $0 \rightarrow 1/\sqrt{2}$, $v \in [0, s]$):

$$2 - u^2 + v^2 = 2 - (\sqrt{2} - s)^2 + v^2 = 2s\sqrt{2} - s^2 + v^2.$$

The inner v -integral: $\int_0^s \frac{2 dv}{2\sqrt{2}s - s^2 + v^2}.$

Writing $b^2 = s(2\sqrt{2} - s)$: $\frac{2}{b} \arctan\left(\frac{v}{b}\right) \Big|_0^s = \frac{2}{b} \arctan\left(\frac{s}{b}\right).$

Now $s/b = s/\sqrt{s(2\sqrt{2} - s)} = \sqrt{s/(2\sqrt{2} - s)}$. With $s = \sqrt{2} \sin^2 \psi$ one finds $\arctan(s/b) = \psi$ and after substitution the outer integral over s yields $\pi^2/8$ as well.

Conclusion.

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \iint_{(0,1)^2} \frac{dx dy}{1 - xy} = 2 \left(\frac{\pi^2}{8} + \frac{\pi^2}{8} \right) = \frac{\pi^2}{4} \cdot \frac{1}{1}$$

Hmm. Let us restate the calculation without error. The correct accounting is:

$$\iint_{(0,1)^2} \frac{dx dy}{1 - xy} = \iint_{S^+ \cup S^-} \frac{2 du dv}{2 - u^2 + v^2} = 2 \iint_{S^+} \frac{2 du dv}{2 - u^2 + v^2} = 2(I_{T_1} + I_{T_2}).$$

where

$$I_{T_1} = \int_0^{1/\sqrt{2}} \frac{2}{\sqrt{2-u^2}} \arctan\left(\frac{u}{\sqrt{2-u^2}}\right) du.$$

With $u = \sqrt{2} \sin \phi$: $\arctan(\tan \phi) = \phi$, $\sqrt{2-u^2} = \sqrt{2} \cos \phi$, $du = \sqrt{2} \cos \phi d\phi$:

$$I_{T_1} = \int_0^{\pi/4} \frac{2}{\sqrt{2} \cos \phi} \cdot \phi \cdot \sqrt{2} \cos \phi d\phi = 2 \int_0^{\pi/4} \phi d\phi = 2 \cdot \frac{\pi^2}{32} = \frac{\pi^2}{16}.$$

By the symmetry of the integrand under $u \mapsto \sqrt{2} - u$ (which maps T_2 to T_1) and the substitution computation above, $I_{T_2} = I_{T_1} = \pi^2/16$. Therefore:

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = 2 \left(\frac{\pi^2}{16} + \frac{\pi^2}{16} \right) = 2 \cdot \frac{\pi^2}{8} = \frac{\pi^2}{4}.$$

This gives $\pi^2/4$, not $\pi^2/6$. The error is in Step 1:

$$\iint_{(0,1)^2} (xy)^n dx dy = \left(\int_0^1 x^n dx \right)^2 = \frac{1}{(n+1)^2},$$

so $\sum_{n=0}^{\infty} \frac{1}{(n+1)^2} = \sum_{n=1}^{\infty} \frac{1}{n^2}$ is correct. The error must be in the rotation.

Let us recheck the symmetry: the full rotated domain is the square with vertices $(0,0)$, $(1/\sqrt{2}, -1/\sqrt{2})$, $(\sqrt{2}, 0)$, $(1/\sqrt{2}, 1/\sqrt{2})$. In $v \mapsto -v$ symmetry the two halves are S^+ ($v \geq 0$) and S^- ($v \leq 0$), and the substitution gives:

$$\iint \frac{2 du dv}{2 - u^2 + v^2}$$

over the whole square. The square is not $S^+ \cup S^-$ with the previous description; let us directly use:

$$\begin{aligned} I &= 4 \int_0^{1/\sqrt{2}} \int_0^u \frac{2 dv du}{2 - u^2 + v^2} + 4 \int_{1/\sqrt{2}}^{\sqrt{2}} \int_0^{\sqrt{2}-u} \frac{2 dv du}{2 - u^2 + v^2} \\ &= 4(I_{T_1} + I_{T_2}). \end{aligned}$$

(Factor 4: factor 2 for $v \mapsto -v$ symmetry, times 2 for the factor in the numerator $2 dv$.) With $I_{T_1} = \pi^2/16$ as above and $I_{T_2} = \pi^2/16$:

$$I = 4 \left(\frac{\pi^2}{16} + \frac{\pi^2}{16} \right) = \frac{\pi^2}{4}.$$

This contradicts the known result. The resolution: $I_{T_2} \neq I_{T_1}$. Let us recompute I_{T_2} carefully.

On T_2 : $u \in [1/\sqrt{2}, \sqrt{2}]$, $v \in [0, \sqrt{2} - u]$. Note $2 - u^2$ is *negative* for $u > \sqrt{2}$, but on T_2 we have $u \leq \sqrt{2}$ so $2 - u^2 \geq 0$. At $u = 1/\sqrt{2}$: $2 - u^2 = 3/2 > 0$. At $u = \sqrt{2}$: $2 - u^2 = 0$.

$$\text{Inner integral: } \int_0^{\sqrt{2}-u} \frac{2 dv}{(2 - u^2) + v^2} = \frac{2}{\sqrt{2 - u^2}} \arctan \left(\frac{\sqrt{2 - u^2}}{\sqrt{2 - u^2}} \right).$$

$$\text{Note } \frac{\sqrt{2 - u^2}}{\sqrt{2 - u^2}} = \frac{\sqrt{2 - u^2}}{\sqrt{(\sqrt{2} - u)(\sqrt{2} + u)}} = \sqrt{\frac{\sqrt{2} - u}{\sqrt{2} + u}}.$$

With $u = \sqrt{2} \cos(2\alpha)$, $\alpha \in [0, \pi/4]$: $\sqrt{2} - u = \sqrt{2}(1 - \cos 2\alpha) = 2\sqrt{2} \sin^2 \alpha$, $\sqrt{2} + u = \sqrt{2}(1 + \cos 2\alpha) = 2\sqrt{2} \cos^2 \alpha$, $2 - u^2 = 2 \sin(2\alpha) \sqrt{2} \dots$ — this substitution becomes complicated.

A cleaner approach uses $u = \sqrt{2} \sin \phi$ with $\phi \in [\pi/4, \pi/2]$ for T_2 : $\sqrt{2} - u = \sqrt{2}(1 - \sin \phi)$, $\sqrt{2 - u^2} = \sqrt{2} \cos \phi$, $\frac{\sqrt{2} - u}{\sqrt{2 - u^2}} = \frac{1 - \sin \phi}{\cos \phi} = \tan(\pi/4 - \phi/2)$ (using the half-angle identity).

$$\text{So } \arctan \left(\frac{\sqrt{2} - u}{\sqrt{2 - u^2}} \right) = \frac{\pi}{4} - \frac{\phi}{2}.$$

The outer integral on T_2 , with $du = \sqrt{2} \cos \phi d\phi$:

$$\begin{aligned} I_{T_2} &= \int_{\pi/4}^{\pi/2} \frac{2}{\sqrt{2} \cos \phi} \left(\frac{\pi}{4} - \frac{\phi}{2} \right) \sqrt{2} \cos \phi d\phi = 2 \int_{\pi/4}^{\pi/2} \left(\frac{\pi}{4} - \frac{\phi}{2} \right) d\phi \\ &= 2 \left[\frac{\pi}{4} \phi - \frac{\phi^2}{4} \right]_{\pi/4}^{\pi/2} = 2 \left[\left(\frac{\pi^2}{8} - \frac{\pi^2}{16} \right) - \left(\frac{\pi^2}{16} - \frac{\pi^2}{64} \right) \right] = 2 \left[\frac{\pi^2}{16} - \frac{\pi^2}{16} + \frac{\pi^2}{64} \right] = \frac{\pi^2}{32}. \end{aligned}$$

Therefore:

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = 4(I_{T_1} + I_{T_2}) = 4 \left(\frac{\pi^2}{16} + \frac{\pi^2}{32} \right) = 4 \cdot \frac{3\pi^2}{32} = \frac{3\pi^2}{8}.$$

This is still not $\pi^2/6$. We must recheck the symmetry $v \mapsto -v$ factor. The integrand $2/(2 - u^2 + v^2)$ is even in v , so the integral over the full rotated square (which is symmetric in

v) equals $2\times$ the integral over $v \geq 0$. The full square in (u, v) has $v \in [-u, u]$ for $u \leq 1/\sqrt{2}$ and $v \in [-(\sqrt{2}-u), \sqrt{2}-u]$ for $u \geq 1/\sqrt{2}$. So the factor from $v \rightarrow -v$ is already accounted for within $T_1 \cup T_2$ (each triangle covers $v \in [0, \cdot]$ and its mirror). Hence *no extra factor of 2* from the v symmetry — the factor 4 above is wrong; it should be 2:

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = 2(I_{T_1} + I_{T_2}) = 2\left(\frac{\pi^2}{16} + \frac{\pi^2}{32}\right) = \frac{\pi^2}{8} + \frac{\pi^2}{16} = \frac{3\pi^2}{16}.$$

Still not $\pi^2/6$. There is an error somewhere in the domain decomposition. Let us restart from scratch with a clean, unambiguous decomposition following Apostol's presentation.

Clean derivation (following Apostol).

The rotation $x = (u+v)/\sqrt{2}$, $y = (u-v)/\sqrt{2}$ maps the unit square $[0, 1]^2$ to the rotated square with vertices $(0, 0)$, $(1/\sqrt{2}, 1/\sqrt{2})$, $(\sqrt{2}, 0)$, $(1/\sqrt{2}, -1/\sqrt{2})$. Since the integrand is even in v , we integrate over $v \geq 0$ and double:

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = 2 \iint_{R^+} \frac{2 \, du \, dv}{2 - u^2 + v^2}, \quad (1)$$

where $R^+ = \{(u, v) : u \geq 0, v \geq 0, u+v \leq \sqrt{2}, u \geq v\}$ (upper-half of the rotated square, i.e. $v \in [0, u]$, $u+v \leq \sqrt{2}$).

R^+ again splits at $u = 1/\sqrt{2}$: $T_1^+ = \{v \in [0, u], u \in [0, 1/\sqrt{2}]\}$ and $T_2^+ = \{v \in [0, \sqrt{2}-u], u \in [1/\sqrt{2}, \sqrt{2}]\}$.

From above: $I_{T_1^+} = \pi^2/16$ and $I_{T_2^+} = \pi^2/32$.

Substituting into (1):

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = 2 \cdot 2(I_{T_1^+} + I_{T_2^+}) = 4\left(\frac{\pi^2}{16} + \frac{\pi^2}{32}\right) = \frac{\pi^2}{4} + \frac{\pi^2}{8} = \frac{3\pi^2}{8}.$$

This is incorrect. The standard proof gives $\pi^2/6$. Our error must be in claiming the $v \rightarrow -v$ symmetry gives a factor of 2: in fact the rotated square has $|v| \leq u$ for $u \leq 1/\sqrt{2}$ and $|v| \leq \sqrt{2}-u$ for $u \geq 1/\sqrt{2}$, so integrating $v \geq 0$ and doubling is correct and gives a factor of exactly 2 applied to $\iint_{R^+} 2/(2 - u^2 + v^2) \, du \, dv$. So we need:

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = 2(I_{T_1^+} + I_{T_2^+}) = 2\left(\frac{\pi^2}{16} + \frac{\pi^2}{32}\right) = \frac{\pi^2}{8} + \frac{\pi^2}{16} = \frac{3\pi^2}{16}.$$

This is still wrong. Let us recheck $I_{T_1^+}$.

$$I_{T_1^+} = \int_0^{1/\sqrt{2}} \int_0^u \frac{2 \, dv \, du}{2 - u^2 + v^2}.$$

Inner integral: $\frac{2}{\sqrt{2}-u^2} \arctan\left(\frac{u}{\sqrt{2}-u^2}\right)$.

With $u = \sqrt{2} \sin \phi$, $\phi \in [0, \pi/4]$: $\sqrt{2}-u^2 = \sqrt{2} \cos \phi$, $u/\sqrt{2}-u^2 = \tan \phi$, $du = \sqrt{2} \cos \phi \, d\phi$:

$$I_{T_1^+} = \int_0^{\pi/4} \frac{2}{\sqrt{2} \cos \phi} \cdot \phi \cdot \sqrt{2} \cos \phi \, d\phi = 2 \int_0^{\pi/4} \phi \, d\phi = 2 \cdot \frac{(\pi/4)^2}{2} = \frac{\pi^2}{16}.$$

This is correct.

Let us recheck $I_{T_2^+}$ using a cleaner substitution. With $u = \sqrt{2} \sin \phi$, $\phi \in [\pi/4, \pi/2]$: $\sqrt{2} - u = \sqrt{2}(1 - \sin \phi)$, $\sqrt{2} - u^2 = \sqrt{2} \cos \phi$, $du = \sqrt{2} \cos \phi d\phi$.

Inner integral upper limit: $\sqrt{2} - u = \sqrt{2}(1 - \sin \phi)$.

$$\int_0^{\sqrt{2}(1-\sin \phi)} \frac{2 dv}{2 - u^2 + v^2} = \frac{2}{\sqrt{2} \cos \phi} \arctan \left(\frac{\sqrt{2}(1 - \sin \phi)}{\sqrt{2} \cos \phi} \right) = \frac{\sqrt{2}}{\cos \phi} \arctan \left(\frac{1 - \sin \phi}{\cos \phi} \right).$$

Using the identity $\frac{1 - \sin \phi}{\cos \phi} = \tan \left(\frac{\pi}{4} - \frac{\phi}{2} \right)$:

$$= \frac{\sqrt{2}}{\cos \phi} \left(\frac{\pi}{4} - \frac{\phi}{2} \right).$$

Outer integral:

$$I_{T_2^+} = \int_{\pi/4}^{\pi/2} \frac{\sqrt{2}}{\cos \phi} \left(\frac{\pi}{4} - \frac{\phi}{2} \right) \sqrt{2} \cos \phi d\phi = 2 \int_{\pi/4}^{\pi/2} \left(\frac{\pi}{4} - \frac{\phi}{2} \right) d\phi.$$

Let $\psi = \pi/2 - \phi$ (so $\phi = \pi/2 - \psi$, $d\phi = -d\psi$, limits $\pi/4 \rightarrow 0$):

$$= 2 \int_0^{\pi/4} \left(\frac{\pi}{4} - \frac{\pi/2 - \psi}{2} \right) d\psi = 2 \int_0^{\pi/4} \frac{\psi}{2} d\psi = \int_0^{\pi/4} \psi d\psi = \frac{(\pi/4)^2}{2} = \frac{\pi^2}{32}.$$

So indeed $I_{T_2^+} = \pi^2/32$ and:

$$I_{T_1^+} + I_{T_2^+} = \frac{\pi^2}{16} + \frac{\pi^2}{32} = \frac{3\pi^2}{32}.$$

The total integral over the rotated square (using v -symmetry):

$$\iint_{\text{square}} \frac{2 du dv}{2 - u^2 + v^2} = 2(I_{T_1^+} + I_{T_2^+}) = \frac{3\pi^2}{16}.$$

But we need this to equal $\sum 1/n^2$. There is a discrepancy with $\pi^2/6$: $3\pi^2/16 \neq \pi^2/6$. We must have an error in the rotation step. Let us recheck.

The rotation sends $(x, y) \rightarrow (u, v)$ with $x = (u + v)/\sqrt{2}$, $y = (u - v)/\sqrt{2}$. Then $xy = \frac{u^2 - v^2}{2}$ and $1 - xy = 1 - \frac{u^2 - v^2}{2} = \frac{2 - u^2 + v^2}{2}$. So the integrand transforms as:

$$\frac{1}{1 - xy} = \frac{2}{2 - u^2 + v^2}.$$

The Jacobian of the inverse map $(u, v) \rightarrow (x, y)$ is 1 (orthogonal transformation), so $dx dy = du dv$. This is all correct.

The rotated unit square has corners: $(x, y) = (0, 0) \rightarrow (u, v) = (0, 0)$; $(x, y) = (1, 0) \rightarrow (u, v) = (1/\sqrt{2}, -1/\sqrt{2})$; $(x, y) = (0, 1) \rightarrow (u, v) = (1/\sqrt{2}, 1/\sqrt{2})$; $(x, y) = (1, 1) \rightarrow (u, v) = (\sqrt{2}, 0)$.

So the rotated square in (u, v) is indeed the rhombus with those four vertices. By v -symmetry, the integral over the upper half ($v \geq 0$) equals half the total. The upper half is:

$$\{u \geq v \geq 0\} \cap \{u \leq \sqrt{2} - v \text{ or } u \leq 1/\sqrt{2}\}.$$

More precisely, the upper half of the rhombus is the triangle with vertices $(0, 0)$, $(1/\sqrt{2}, 1/\sqrt{2})$, $(\sqrt{2}, 0)$, bounded by $v \leq u$ and $v \leq \sqrt{2} - u$. This is $R^+ = \{v \in [0, \min(u, \sqrt{2} - u)]\}$.

For $u \in [0, 1/\sqrt{2}]$: $\min(u, \sqrt{2} - u) = u$, so $v \in [0, u]$ (T_1^+). For $u \in [1/\sqrt{2}, \sqrt{2}]$: $\min(u, \sqrt{2} - u) = \sqrt{2} - u$, so $v \in [0, \sqrt{2} - u]$ (T_2^+). This matches what we had.

The total integral is:

$$\iint_{\text{rhombus}} \frac{2}{2 - u^2 + v^2} du dv = 2 \iint_{R^+} \frac{2}{2 - u^2 + v^2} du dv = 4(I_{T_1^+} + I_{T_2^+}) = \frac{3\pi^2}{8}.$$

We are getting $3\pi^2/8 = 3\pi^2/8 \approx 3.70$, but $\pi^2/6 \approx 1.645$. Clearly $3\pi^2/8 \neq \pi^2/6$. This means we have an algebraic error somewhere. Let us numerically check $\sum_{n=1}^{1000} 1/n^2 \approx 1.6439$ and compute the double integral numerically.

$\iint_{(0,1)^2} \frac{dx dy}{1-xy}$: near $(1, 1)$ the integrand diverges. The integral is improper but convergent: $\int_0^1 \int_0^1 (xy)^n dx dy = 1/(n+1)^2$ and $\sum_{n=0}^{\infty} 1/(n+1)^2 = \pi^2/6$. So $\pi^2/6$ is correct for the double integral.

The discrepancy must be in $I_{T_1^+}$. Let us recheck numerically. $I_{T_1^+} = 2 \int_0^{\pi/4} \phi d\phi = 2 \cdot \pi^2/32 = \pi^2/16 \approx 0.617$. $I_{T_2^+} = \pi^2/32 \approx 0.308$. $4(I_{T_1^+} + I_{T_2^+}) = 4(\pi^2/16 + \pi^2/32) = 4 \cdot 3\pi^2/32 = 3\pi^2/8 \approx 3.70$. But $\pi^2/6 \approx 1.645$. Factor of ≈ 2.25 off.

The issue: the rhombus has $v \in [-u, u]$, not just $v \in [0, u]$, and similarly for T_2 . So when we say ‘‘upper half’’ and double, we must be integrating v from $-\min(u, \sqrt{2} - u)$ to $+\min(u, \sqrt{2} - u)$, which gives exactly twice the $v \geq 0$ part. So a factor of 2 from v -symmetry is correct.

Let us try a direct numerical estimate of $I_{T_1^+}$: At $u = 0.5$: inner integral $= \int_0^{0.5} \frac{2 dv}{2 - 0.25 + v^2} = \int_0^{0.5} \frac{2 dv}{1.75 + v^2} = \frac{2}{\sqrt{1.75}} \arctan(0.5/\sqrt{1.75}) \approx \frac{2}{1.3229} \arctan(0.3780) \approx 1.5119 \cdot 0.3614 \approx 0.546$. Outer integral $\approx \int_0^{0.707} 0.546 du \approx 0.386$. True $I_{T_1^+} = \pi^2/16 \approx 0.617$. The discrepancy suggests the inner integral formula is wrong.

Recheck: $\int_0^u \frac{2 dv}{2 - u^2 + v^2}$. At $u = 0.5$: $2 - u^2 = 1.75$, upper limit $v = 0.5$, so $\frac{2}{\sqrt{1.75}} \arctan(0.5/\sqrt{1.75}) \approx 0.546$. OK.

So the outer integral is $\int_0^{1/\sqrt{2}} f(u) du$ where $f(u) = \frac{2}{\sqrt{2-u^2}} \arctan(u/\sqrt{2-u^2})$.

Substitution $u = \sqrt{2} \sin \phi$, $du = \sqrt{2} \cos \phi d\phi$, $\sqrt{2-u^2} = \sqrt{2} \cos \phi$, limit $u = 1/\sqrt{2} \Leftrightarrow \sin \phi = 1/2 \Leftrightarrow \phi = \pi/6$ (not $\pi/4$!).

We had an error in the upper limit. $u = 1/\sqrt{2}$ and $u = \sqrt{2} \sin \phi$ gives $\sin \phi = \frac{1/\sqrt{2}}{\sqrt{2}} = \frac{1}{2}$, so $\phi = \pi/6$, not $\pi/4$.

Correcting:

$$I_{T_1^+} = 2 \int_0^{\pi/6} \phi d\phi = 2 \cdot \frac{(\pi/6)^2}{2} = \frac{\pi^2}{36}.$$

Now $I_{T_2^+}$: on T_2^+ , $u \in [1/\sqrt{2}, \sqrt{2}]$ corresponds to $\phi \in [\pi/6, \pi/2]$.

$$I_{T_2^+} = 2 \int_{\pi/6}^{\pi/2} \left(\frac{\pi}{4} - \frac{\phi}{2} \right) d\phi.$$

Let $\psi = \pi/2 - \phi$, $d\phi = -d\psi$, limits $\pi/3 \rightarrow 0$:

$$= 2 \int_0^{\pi/3} \frac{\psi}{2} d\psi = \int_0^{\pi/3} \psi d\psi = \frac{(\pi/3)^2}{2} = \frac{\pi^2}{18}.$$

Total:

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = 4(I_{T_1^+} + I_{T_2^+}) = 4\left(\frac{\pi^2}{36} + \frac{\pi^2}{18}\right) = 4 \cdot \frac{\pi^2}{12} = \frac{\pi^2}{3}.$$

Still not $\pi^2/6$. We are off by a factor of 2.

The remaining error: the v -symmetry factor. When we say “upper half and double”, we get: $\iint_{\text{rhombus}} = 2 \iint_{R^+}$. And $\iint_{R^+} = I_{T_1^+} + I_{T_2^+}$ (integrating $2/(2 - u^2 + v^2)$ over $v \in [0, \min(u, \sqrt{2} - u)]$). So total = $2(I_{T_1^+} + I_{T_2^+})$, not $4(I_{T_1^+} + I_{T_2^+})$.

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = 2\left(\frac{\pi^2}{36} + \frac{\pi^2}{18}\right) = 2 \cdot \frac{\pi^2}{12} = \frac{\pi^2}{6}. \quad \checkmark$$

Summary of the correct proof.

Step 1. $\sum_{n=1}^{\infty} \frac{1}{n^2} = \iint_{(0,1)^2} \frac{dx dy}{1 - xy}.$

Step 2. Rotation $u = (x + y)/\sqrt{2}$, $v = (x - y)/\sqrt{2}$ gives $1 - xy = (2 - u^2 + v^2)/2$ and $dx dy = du dv$. The unit square maps to the rhombus R with vertices $(0, 0)$, $(1/\sqrt{2}, \pm 1/\sqrt{2})$, $(\sqrt{2}, 0)$:

$$\iint_{(0,1)^2} \frac{dx dy}{1 - xy} = \iint_R \frac{2 du dv}{2 - u^2 + v^2}.$$

Step 3. By $v \mapsto -v$ symmetry, integrating over $R^+ = R \cap \{v \geq 0\}$ and doubling:

$$= 2 \iint_{R^+} \frac{2 du dv}{2 - u^2 + v^2} = 2(I_{T_1^+} + I_{T_2^+}).$$

Step 4. R^+ splits at $u = 1/\sqrt{2}$. On T_1^+ : $u \in [0, 1/\sqrt{2}]$, $v \in [0, u]$. Using $u = \sqrt{2} \sin \phi$, $\phi \in [0, \pi/6]$:

$$I_{T_1^+} = \int_0^{1/\sqrt{2}} \frac{2}{\sqrt{2 - u^2}} \arctan\left(\frac{u}{\sqrt{2 - u^2}}\right) du = 2 \int_0^{\pi/6} \phi d\phi = \frac{\pi^2}{36}.$$

Step 5. On T_2^+ : $u \in [1/\sqrt{2}, \sqrt{2}]$, $v \in [0, \sqrt{2} - u]$. Using $u = \sqrt{2} \sin \phi$, $\phi \in [\pi/6, \pi/2]$, and $\arctan\left(\frac{\sqrt{2} - u}{\sqrt{2 - u^2}}\right) = \frac{\pi}{4} - \frac{\phi}{2}$:

$$I_{T_2^+} = 2 \int_{\pi/6}^{\pi/2} \left(\frac{\pi}{4} - \frac{\phi}{2}\right) d\phi = \int_0^{\pi/3} \psi d\psi = \frac{\pi^2}{18}.$$

Step 6.

$$\boxed{\sum_{n=1}^{\infty} \frac{1}{n^2} = 2\left(\frac{\pi^2}{36} + \frac{\pi^2}{18}\right) = 2 \cdot \frac{3\pi^2}{36} = \frac{\pi^2}{6}.} \quad \square$$

Conclusion: $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}. \quad \square$

This result appears in the partition function of phonons at low temperature (Debye model), in the calculation of the Casimir effect, and in the study of the Riemann zeta function $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$, which encodes the distribution of prime numbers.

4.3 Volume of the n -Dimensional Ball

Let $B_n(R) = \{(x_1, \dots, x_n) \in \mathbb{R}^n : x_1^2 + \dots + x_n^2 \leq R^2\}$. Define $V_n(R) = \text{Vol}(B_n(R))$. By dimensional analysis, $V_n(R) = \omega_n R^n$ where $\omega_n = V_n(1)$.

Recursion. Slicing $B_n(R)$ at height $x_n = t$:

$$V_n(R) = \int_{-R}^R V_{n-1}(\sqrt{R^2 - t^2}) dt = \omega_{n-1} \int_{-R}^R (R^2 - t^2)^{(n-1)/2} dt.$$

With $t = R \sin \psi$: the integral evaluates via the Beta function $B(a, b) = \int_0^1 u^{a-1} (1-u)^{b-1} du = \Gamma(a)\Gamma(b)/\Gamma(a+b)$, giving the recursion:

$$\omega_n = \frac{\pi}{n/2} \omega_{n-2}, \quad \omega_1 = 2, \quad \omega_2 = \pi.$$

Solving:

$$V_n(R) = \frac{\pi^{n/2}}{\Gamma(n/2 + 1)} R^n.$$

Specific values.

n	1	2	3	4	n
$V_n(R)$	$2R$	πR^2	$\frac{4}{3}\pi R^3$	$\frac{\pi^2}{2} R^4$	$\frac{\pi^{n/2}}{\Gamma(n/2+1)} R^n$

Note: $V_n(1) \rightarrow 0$ as $n \rightarrow \infty$ — the unit ball in high dimensions has vanishingly small volume relative to the unit cube. Almost all the volume of a high-dimensional ball is concentrated near its surface, a fact central to statistical mechanics and machine learning.

4.4 Green's Theorem, Stokes' Theorem, and the Divergence Theorem

We computed line integrals in Meeting 2 and found that for exact forms, $\oint_\gamma df = 0$. This is a special case of a much deeper result.

Key operators. For $\mathbf{F} = (F_1, F_2, F_3)$ and scalar f , we define:

$$\text{Gradient: } \nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right)$$

$$\text{Divergence: } \nabla \cdot \mathbf{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$$

$$\text{Curl: } \nabla \times \mathbf{F} = \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}, \frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x}, \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right)$$

$$\text{Laplacian: } \Delta f = \nabla \cdot (\nabla f) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$$

Green's theorem (2D). For a region $D \subseteq \mathbb{R}^2$ with smooth boundary ∂D :

$$\oint_{\partial D} P dx + Q dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy.$$

The left side is the line integral around the boundary; the right side integrates the “curl” of (P, Q) over the interior.

When $\partial Q/\partial x = \partial P/\partial y$ (exactness condition!), the right side is zero, recovering $\oint df = 0$ for exact forms. Green’s theorem is the *explanation* of why exact forms integrate to zero on closed loops: the integrand of the area integral vanishes identically.

Classical Stokes’ theorem (3D surface). For an oriented surface S with boundary curve ∂S :

$$\oint_{\partial S} \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}.$$

A vector field $\mathbf{F}$ is conservative (i.e., $\mathbf{F} = \nabla f$) if and only if $\nabla \times \mathbf{F} = 0$. By Stokes, this means $\oint_{\partial S} \mathbf{F} \cdot d\mathbf{r} = 0$ for every closed curve, exactly as we proved for exact forms.

Divergence theorem (Gauss’s theorem). For a volume Ω with boundary surface $\partial\Omega$:

$$\iint_{\partial\Omega} \mathbf{F} \cdot d\mathbf{S} = \iiint_{\Omega} \nabla \cdot \mathbf{F} dV.$$

The flux of $\mathbf{F}$ through a closed surface equals the integral of its divergence over the enclosed volume.

Physical content.

- Gauss’s law: $\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$ (the non-exact form $\Omega = \mathbf{r}/r^3 \cdot d\mathbf{S}$ from Meeting 2 corresponds to $\nabla \cdot \mathbf{E} = 0$ away from charges, but the divergence theorem detects the charge at the origin).
- Faraday’s law: $\nabla \times \mathbf{E} = -\partial\mathbf{B}/\partial t$ (a non-zero curl means the field is not conservative, i.e., the form $\mathbf{E} \cdot d\mathbf{r}$ is not exact in time-varying fields).
- Continuity equation: $\partial\rho/\partial t + \nabla \cdot \mathbf{J} = 0$ (the divergence theorem converts local conservation into a surface flux).

The unifying statement. All three theorems are special cases of a single formula on differential forms, known as the **generalised Stokes theorem**:

$$\int_{\partial M} \omega = \int_M d\omega.$$

Here M is any smooth manifold with boundary ∂M , and $d\omega$ is the *exterior derivative* of the form ω . In the language of Meeting 2: the exactness condition $d\omega = 0$ is precisely the condition $\nabla \times \mathbf{F} = 0$ (or $\partial P/\partial y = \partial Q/\partial x$ in 2D). The “measure of non-exactness” $d\omega$ is what gets integrated over the interior when we apply any version of Stokes’ theorem.

Summary

Key formulas

Change of variables:	$\int_V f \, dx = \int_U f(\Phi(u)) \det D\Phi \, du$
Polar area element:	$dx \, dy = r \, dr \, d\theta$
Spherical volume element:	$dx \, dy \, dz = r^2 \sin \theta \, dr \, d\theta \, d\varphi$
Gaussian:	$\int_{-\infty}^{\infty} e^{-x^2} \, dx = \sqrt{\pi}$
Ball volume:	$V_n(R) = \frac{\pi^{n/2}}{\Gamma(n/2 + 1)} R^n$
Green's theorem:	$\oint_{\partial D} P \, dx + Q \, dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$

Looking ahead. The generalised Stokes theorem $\int_{\partial M} \omega = \int_M d\omega$ unifies everything we have done: line integrals of exact forms (Meeting 2), the topology of the punctured plane (winding number as a cohomological invariant), and the divergence and Stokes theorems. The machinery of differential forms and de Rham cohomology gives the complete answer to the guiding question of Meeting 2: a form ω integrates to zero over every closed surface of dimension k if and only if ω is exact, and the obstruction to exactness is measured by the k -th cohomology group $H^k(M)$.